

भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 306: Machine Learning Lab
Practice Assignment 1

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Write a program to create two different synthetic datasets, each having two variables (one dataset having high variance and the other dataset having low variance). Create scatter plots to visualize the relationship between X and Y, for both datasets. Report and interpret the value of covariance and the Pearson's correlation coefficient between them, for both datasets.

2. Write a program to create a synthetic dataset with two variables (having non-linear relation). Create a scatter plot to visualize the relationship between X and Y. Report and interpret the value of covariance and the Pearson's correlation coefficient between them. From the results, report your analysis / findings.

3. Write a program to create a synthetic dataset with two variables for the two following scenarios:

   (a) the independent variable (X) has monotonic values.

   (b) the independent variable (X) has non-monotonic values

   For both scenarios, write programs to determine which correlation coefficient measure will be appropriate and to calculate the values. Also, visualize the shape of the curve obtained for both the scenarios, and report and interpret the results / values of correlation coefficients.

4. (a) Write a program to create a synthetic dataset for car reselling price with three numerical variables as below:

      i. price: Target variable (in dollars/ Rs.).

      ii. mileage: Independent variable.

      iii. Age of the car: Independent variable (can be calculated as ((current year) - (year of manufacture)), where current year is 2025.

   (b) Write a program to calculate the Pearson's correlation coefficients for all the variable pairs and display it. Determine the pairs with highest (positive / negative) correlations and interpret the results.